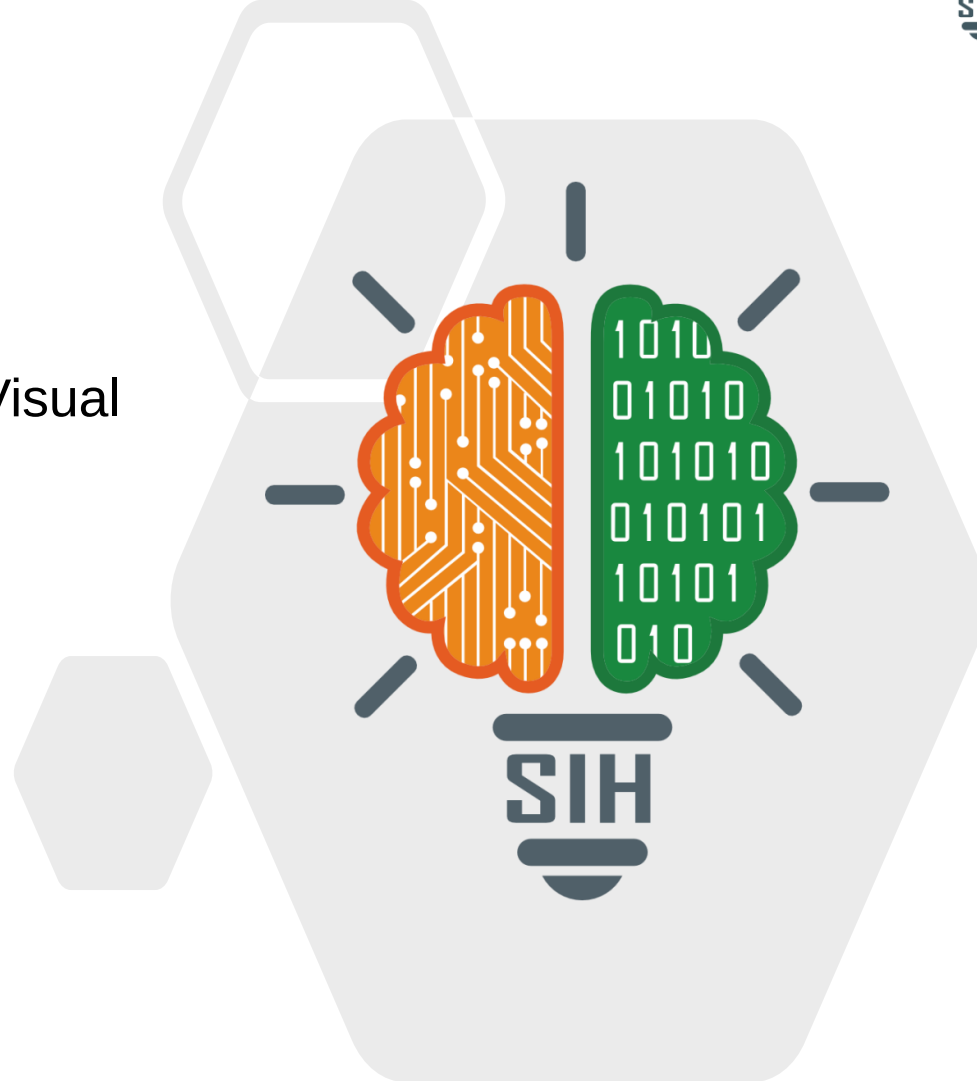


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26171
- **Problem Statement Title** - On-device Visual Perception for Light-weight Browser Agents
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 167230
- **Team Name** - SuperNova!



Privacy-First Browser Agent

SuperNova: a browser agent that sees locally, reasons remotely and acts automatically

1 USER TASK

Goal typed in the popup, e.g. "Find and open the cheapest plan"

2 LOCAL PERCEPTION

DOM tree (≤ 60 elements) plus screenshot; OWL-ViT vision and Tesseract OCR run on-device

3 PII PROTECTION

Regex, OCR and vision detect PII; black-box redaction and [PLACEHOLDERS]

4 SERVER AI

Leak check, then a VLM plans one JSON action from sanitized DOM + redacted image

5 AUTO ACTION

Click / type / scroll / navigate via safety gate; loops up to 10 steps

How it addresses the problem

- Client-side vision on WebGPU: OWL-ViT via Transformers.js / ONNX Runtime Web, WASM fallback
- Demonstrable privacy filter: before/after thumbnails and redaction summary in popup
- Only anonymised data leaves: sanitized DOM + redacted JPEG, re-checked by server
- Server returns UI actions: structured JSON executed on the page by the client
- End-to-end task: multi-step loop with action history and retry on failure
- Latency vs accuracy: DOM first, vision + OCR run in parallel

Innovation and uniqueness

- Three-layer redaction (DOM + OCR + vision) also catches PII drawn on canvas or in images
- India-ready PII: Aadhaar, PAN and +91 numbers alongside email, card and SSN
- Zero-trust server: independent leak check discards any unsafe context
- Redaction-aware planning: placeholders and black boxes preserve layout for the AI
- Safety gate, 10-step cap and kill switch limit agent autonomy
- Bundled local models: works offline, no third-party image upload

What stays local vs. what is sent

STAYS ON DEVICE

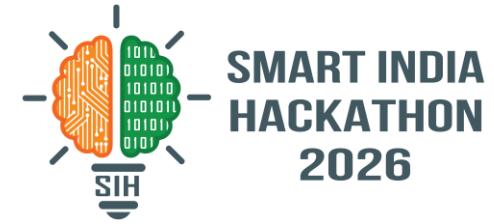
- Raw screenshot and pixels
- Faces, ID cards, passports
- Passwords, emails, phone, Aadhaar, PAN
- Model weights and inference

SENT TO SERVER

- Redacted JPEG (max 1280 px)
- Sanitized DOM: tag, role, text, position
- Task and recent action history

SuperNova!

TECHNICAL APPROACH



ON-DEVICE PRIVACY LAYER (browser)

- 1 Content script: task + page capture
- 2 DOM extraction: ≤ 60 elements, viewport-first, stamped selectors
- 3 OWL-ViT (WebGPU $\rightarrow$ WASM): face, ID card, credit card, passport
- 4 Tesseract OCR + PII regex $\rightarrow$ word-level boxes
- 5 Canvas redaction $\rightarrow$ 1280 px JPEG + sanitized DOM

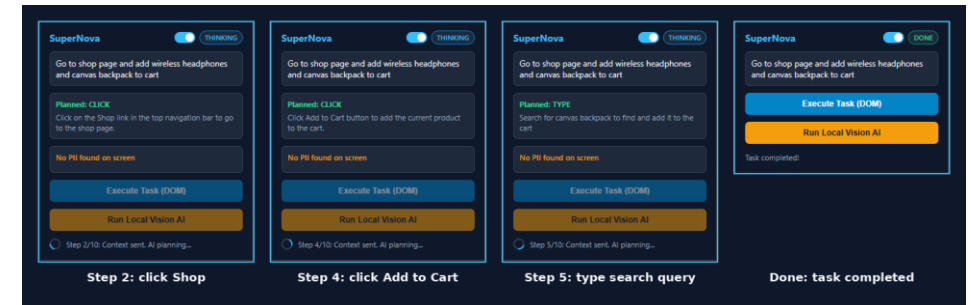
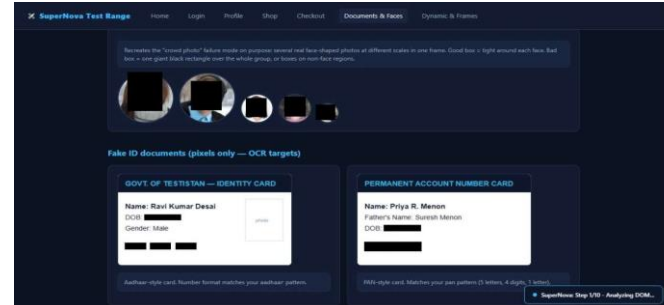
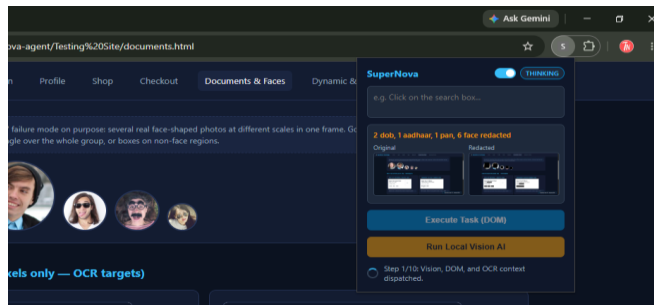
SERVER REASONING LAYER

- WebSocket with origin allowlist (chrome-extension only)
- Leak check: regex re-scan of DOM; discard on any hit
- VLM planner: task + DOM + screenshot + history
- Returns Pydantic-validated JSON action

TECHNOLOGIES USED

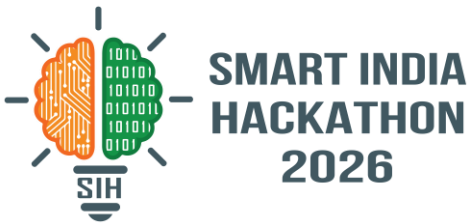
- **Client:** Chrome MV3 (service worker, content script, offscreen document), JavaScript
- **Vision:** Transformers.js + ONNX Runtime Web; OWL-ViT base patch32 (zero-shot, ~ 152 MB); WebGPU with WASM fallback
- **OCR:** Tesseract.js (WASM, English)
- **PII engine:** Regex for email, phone, SSN, card, Aadhaar, PAN; DOM rules for password/email fields
- **Redaction:** Canvas black boxes, JPEG q0.6
- **Server:** Python, FastAPI, WebSocket, Pydantic
- **Planner:** Cloud VLM (Gemini Flash) now; swappable with open-weights VLM
- **Offline:** all local models bundled, no remote fetch

TRUST BOUNDARY $\rightarrow$ redacted JPEG + sanitized DOM $\leftarrow$ JSON action (safety gate, then DOM execution, next step ≤ 10)





FEASIBILITY AND VIABILITY



TECHNICAL

- WebGPU, WASM and ONNX are production-ready in browsers
- Working extension + FastAPI prototype, end to end
- Models bundled locally; runs offline

OPERATIONAL

- Install as an extension; no workflow change
- Popup shows what was redacted, with thumbnails
- One-click on/off toggle

SECURE

- Pixels, passwords and PII never leave the device
- Server treated as untrusted: origin check + leak check
- User confirms risky actions

SIH metric (weight)	How we address it	Measured	Risk	Mitigation
Visual context accuracy (25%)	DOM semantics + redacted screenshot; vision covers canvas/iframes	8/10 tasks done (2 failures = API overload)	Missed PII	Three layers, low detection threshold, server re-scan discards unsafe context
PII recall / precision (20%)	DOM + OCR + vision layers; 0.15 threshold favours recall	26/30 in-scope PII covered (87% recall)	Over-redaction	Word-level boxes only; [PLACEHOLDERS] keep context for the planner
Redaction precision (20%)	Word-level OCR boxes; tight face/ID boxes; placeholders	2 false redaction across 4 pages	Weak devices	WebGPU → WASM fallback; single-thread WASM; models pre-loaded once
Client resources (20%)	Models load once; WASM single-thread fallback; 1280 px cap	≈514→563 MB RAM; CPU peak ≈190% of 1 core	High latency	Parallel OCR + vision; DOM-first flow; downscaled JPEG
End-to-end latency (15%)	OCR and vision in parallel; DOM first; JPEG q0.6	≈15–16 s/step (≈7.4 s on-device)	API overload	Key pool with rotation and backoff; swappable for self-hosted open-weights VLM
			Excess autonomy	Confirmation gate, 10-step cap, kill switch, duplicate-action detection
			Browser support	Chromium MV3 for MVP; Firefox port on roadmap

Pilot: 4 test pages, 30 in-scope PII items, 10 tasks. In scope: email, phone, Aadhaar, PAN, DOB, faces, ID documents; names/gender out of scope (roadmap). Test device: 12th Gen Intel Core i5-12450HX (2.40 GHz), 12 GB RAM, NVIDIA GeForce RTX 2050 (4 GB) + Intel UHD Graphics, Chrome 154.0.8037.58 (64-bit), Windows 11.

WHO BENEFITS

Everyday users

Safer AI-assisted browsing without exposing screens, IDs or passwords

Banking, UPI, e-governance

Aadhaar / PAN-aware protection on sensitive portals

Enterprises, public sector

Privacy-sensitive workflows; data stays under organizational control

Accessibility and assisted use

Automates complex web forms for users who find them hard

BENEFITS

Social

- Privacy by default for AI assistants
- User can see exactly what was redacted
- Supports data-minimization aims of the DPDP Act 2023

Economic

- Lower server, bandwidth and storage load
- Fewer breach and compliance risks
- Works with open-weights models on existing devices

Environmental

- Small redacted payloads instead of full-resolution screens
- Runs on existing hardware; no new devices
- Smaller redacted payloads reduce data sent per step

MEASURABLE VALUE

26/30

in-scope PII covered (4 pages)

≈15 s

per step; ≈7.4 s on-device

36–39%

payload vs raw screenshot (97→62 KB)

8/10

tasks done; 2 failures = API overload

1. SIH 2026 PS 26171 (ISRO)

Defines the architecture and evaluation criteria

<https://www.sih.gov.in/>

2. OWL-ViT: open-vocabulary detection (Minderer et al., 2022)

Basis of our zero-shot face / ID detector

<https://arxiv.org/abs/2205.06230>

3. Xenova/owlvit-base-patch32

ONNX model used for on-device detection

<https://huggingface.co/Xenova/owlvit-base-patch32>

4. Transformers.js

Browser inference with WebGPU / WASM

<https://huggingface.co/docs/transformers.js/>

5. ONNX Runtime Web (WebGPU)

GPU-accelerated in-browser inference

<https://onnxruntime.ai/docs/tutorials/web/ep-webgpu.html>

6. Tesseract.js

In-browser OCR for text drawn as pixels

<https://github.com/naptha/tesseract.js>

7. Chrome Off-screen API

Runs ML workers from an MV3 extension

<https://developer.chrome.com/docs/extensions/reference/api/offscreen>

8. OWASP LLM06: Excessive Agency

Basis for safety gate, step cap and kill switch

<https://genai.owasp.org/llmrisk/llm062025-excessive-agency/>

9. WebVoyager (He et al., 2024)

Web agent that combines screenshots with page text

<https://arxiv.org/abs/2401.13919>

10. Digital Personal Data Protection Act, 2023

Data-minimization context for the design

<https://indiacode.gov.in/act/c058fa9f-eaf0-4ca3-98f1-3443b087bca9/sections>

PROJECT LINKS

<https://github.com/nirbhayc01/SuperNova>

<https://www.youtube.com/watch?v=No98BMGRDnc>